

JEREMY DE LEON

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Education

University of California, Los Angeles

June 2027

Bachelor's of Science, Mechanical Engineering

Los Angeles, CA

Coursework: Strength of Materials, Dynamics, Mechanisms, Manufacturing, Thermodynamics, MATLAB, Python, Computer Aided Design

Activities: Creative Labs, Theta Tau, X1 Robotics, Rocket Project, Pilipinos in Engineering, Pilipino Cultural Night

Work Experience

UCLA Di Carlo Lab

January 2026 – Present

Undergraduate Researcher — CAD & Manufacturing

Los Angeles, CA

- Designed a toolhead exchange mechanism with Fusion for a benchtop robot, specializing in large language model (LLM) integration to miniaturize & automate full bioassay workflows & microfluidic R&D
- Applied GD&T supporting assembly resolving tolerance & alignment between 3D printed & machined stainless steel parts
- Performed CNC, manual milling, lathe, & drilling operations for custom frame & structural components

American Society of Mechanical Engineers (ASME)

Sept 2025 – Present

Mechanical Engineer – Bruin BB-8

Los Angeles, CA

- Designed an internal arm in SolidWorks achieving 3-axis head articulation at a 253mm radial distance for a holonomic spherical robot inspired by BB-8 from Star Wars
- Specified tight tolerances ($\pm 0.25\text{mm}$ inner, $\pm 0.75\text{mm}$ outer diameter) for a press-fit magnet & dowel-spring preload fastening strategy to ensure repeatable performance under dynamic loading
- Performed FEA to validate structural integrity of 6061-T6 aluminum components confirming 1.5–2x factor of safety prior to manufacturing

UCLA Creative Labs

April 2026 – Present

Project Lead – Petalux

Los Angeles, CA

- Spearheaded the development of an interactive servo actuated flower lamp, coordinating the integration between mechanical, electrical & software subteams
- Drafted & iterated Fusion 360 models for an open/close petal mechanism supported by embedded control implementation in Arduino IDE + mobile map controls

UCLA Rocket Project

September 2024 – June 2025

Mechanical Engineer – Hybrid Rocket

Los Angeles, CA

- Designed a 3-section carbon fiber body tube layup with $\pm 45^\circ/0^\circ$ plies and reinforced bolted-joint regions, achieving strong buckling & bending margins
- Manufactured plywood engine centering rings for the rocket airframe, using laser-cut $\frac{1}{4}$ -inch laminations epoxied into $\frac{1}{2}$ -inch structural rings to align the motor tube, reduce thrust misalignment, & support body tube bending loads

Projects

Bionic Hand

Onshape

- Led the development of a bionic hand prototype, designing mechanical assemblies in Onshape and supporting Arduino IDE-based actuation and control integration for a functional assistive-device concept

Automated Pill Dispenser

SolidWorks

- Designed 15-part SolidWorks assembly for a dual-hopper pill dispenser in PLA with DFM constraints applied across variable pill geometries, delivering a $22.5\text{cm} \times 7.5\text{cm}$ footprint optimized for 3D printability

Ant Colony Simulation

MATLAB

- Developed a probabilistic multi-agent pathfinding simulation in MATLAB across 8 modular functions, implementing concentration-weighted pheromone gradient steering, stochastic angle updates drawn from $\mathcal{N}(0, \sigma)$, and linear pheromone decay to enable emergent shortest-path discovery

Skills

CAD & Analysis: SolidWorks, AutoCAD, Autodesk Fusion, Onshape, Revit, ANSYS, FEA, CFD

Engineering Methods: GD&T, FMEA, DFM/DFA, CNC, 3D Printing, Rapid Prototyping

Programming: MATLAB, Python, C++, Git